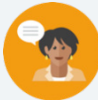


6.12 Multiplying and Dividing Integers With Procedural Fluency

Lesson Introduction

 Benchmark	MA.6.NSO.4.2 Apply and extend previous understandings of operations with whole numbers to multiply and divide integers with procedural fluency.		 Learning Target	I can multiply and divide integers with procedural fluency.
 Benchmark Coherence	Building On		Working Toward	
	MA.5.NSO.2 Add, subtract, multiply, and divide multi-digit numbers.		MA.7.NSO.2.2 Add, subtract, multiply, and divide rational numbers with procedural fluency. MA.7.NSO.2.3 Solve real-world problems involving any of the four operations with rational numbers.	
 Essential Questions	- How are the rules for multiplying more than two factors similar to the rules for multiplying two factors? - How are multiplication and division related?			
 Lesson Narrative	In this lesson, students discover that when multiplying or dividing integers, the sign will be positive if there is an even number of negative factors and negative if there is an odd number of negative factors. Students develop procedural fluency through working several practice problems.			
 Component Summary	6.12.1 Warm-Up (~5-10 min.)	Students select negative expressions from a given number line with numbers and variables plotted (<i>SE p. 288</i>)	6.12.4 Guided Practice (5-10 min.)	Practice finding products and quotients of integers (<i>SE p. 289</i>)
	6.12.2 Collaboration (10 min.)	Collaboratively multiply multiple integers and identify patterns with signs (<i>SE p. 288</i>)	6.12.5 Your Turn! (10 min.)	Additional practice finding products and quotients of integers (<i>SE p. 290</i>)
	6.12.3 Activity (10 min.)	Practice multiplying integers by playing Integer War (<i>SE p. 289</i>)	6.12.6 Wrap-Up (~5 min.)	Demonstrate understanding of lesson learning goals (<i>SE p. 290</i>)
 Resources	Glossary	Materials	Additional Preparation	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Integer - Expression	One deck of cards per student pair	(Optional) Prepare and print sheets for students to record their work for the card game. - Prepare decks of cards by removing face cards or ask students to do this before beginning the game. - Prepare student grouping for 6.12.3 Activity.	

 Teacher Tips	Common Misconceptions • Students may struggle to reason about the absolute value of a and b on the number line in the 6.12.1 Warm-Up. • Students may struggle multiplying more than two integer factors. • Students may struggle accurately multiplying and dividing two- and three-digit numbers.	Things to Consider • Consider creating and displaying an anchor chart with the rules for multiplying multiple factors.
	Literacy and Language Routines Think-Pair-Share 6.12.2 Collaboration provides students the opportunity to discuss their work and answers with a partner. These discussions will help students gain confidence and understanding.	Mathematical Thinking and Reasoning (MTR) Standard MA.K12.MTR.3.1: Mathematical Fluency 6.12.2 Collaboration and 6.12.5 Your Turn! provide opportunities for students to apply what they have learned about multiplying integers to expressions with more than two factors. Encourage students to make connections from what they learned about multiplying two integers to determine the sign of the product of multiple numbers.
	 Differentiate Instruction Integer War 6.12.3 Activity provides a unique, engaging, and collaborative opportunity to increase fluency in integer operations. Consider your students when planning groups, using data and informal observations from the previous lessons to inform pairings as well as the method in which students are engaging. • Pair students by their current understanding or fluency with multiplying integers for “even” pairings. • Alternatively, rather than focusing on speed, one student in each pair is negative and one is positive, or one student is less than 40 and one student is greater than 40. The product determines who wins the cards rather than the speed of who finishes first. • Groups of three can be formed to increase the difficulty for some groups, if appropriate.	
Lesson Notes:		

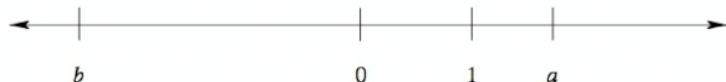
6.12.1 Warm-Up: Thinking Algebraically

Component Overview: Students work independently to use variables and numbers plotted on a number line to identify expressions that will have negative values. After explaining their thinking independently, they can work with a partner to compare expressions and explanations, providing an opportunity to refine or revise their thinking.



6.12.1 Warm-Up: Thinking Algebraically

A number line is shown. The numbers 0 and 1 are marked on the line, along with two other numbers, a and b .



1. Circle one expression from those shown below that will have a negative value.

$a - 1$	$a - 2$	$a + b$	$a - b$	$ab + 1$
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2. Explain how you know the expression selected will have a negative value based on the number line.

Answers will vary. Since the distance between 0 and 1 is greater than the distance between 1 and a , I know that $a < 2$, so $a - 2 < 0$. I see that b is farther away from 0 than a is. Therefore, b has a greater absolute value than a , so $a + b$ will be negative. Since the absolute values of a and b are both greater than 1, ab will be less than -1 , so $ab + 1$ will still be negative.

3. Find a classmate who chose a different expression that will have a negative value.

Record their expression and explanation for why they chose it in the table. Then, have them initial that you recorded it correctly. *You are the only person who should write in the first two columns of the table.*

Answers will vary.

Classmate's Selected Expression	Classmate's Explanation for Why the Value is Negative	Initials



Students may benefit from sentence stems, such as “I know the expression _____ will have a negative value, because _____.”

6.12.2 Collaboration: Multiplying Multiple Integers

Component Overview: Students work with a partner to discover patterns in the signs when multiplying multiple integers. Students discover that the sign will be positive if there are an even number of negative factors and negative if there are an odd number of negative factors. Debrief with a whole-class discussion to consolidate student understanding.



6.12.2 Collaboration: Multiplying Multiple Integers

1. Work with your partner to find the product of each expression.

a. $5(-9)$ -45	b. $-6(-4)$ 24	c. $0(-3)$ 0
d. $(-1)(-1)(3)$ 3	e. $4 \cdot (-2) \cdot (5)$ -40	f. $-2(-2)(-2)(-2)$ 16
g. $(-1) \times (-2) \times 3 \times (-2) \times (-1) \times 2$ 24		

2. In expressions d-g, more than two factors are being multiplied.

Discuss with your partner how the rules you developed previously for multiplying two integers could be extended for multiplying more than two integers.

Write at least one rule that would apply in this situation.

Answers will vary.

$$\begin{array}{c} \text{Red circle with } - \times \text{Red circle with } - = \text{Yellow circle with } + \\ \text{Red circle with } - \times \text{Yellow circle with } + = \text{Red circle with } - \end{array}$$

Multiplying an odd number of negative factors will result in a negative product.

Multiplying an even number of negative factors will result in a positive product.



Consider providing sentence starters for question 2. For example, “The sign will be positive if...” and “The sign will be negative if...”

6.12.3 Activity: Integer War – Multiplication Edition

Component Overview: Students use a deck of playing cards to play Integer War in pairs. Both students flip over a card, and the first student to state the product wins the round and collects both cards. Before beginning, model a few rounds for students. Consider providing sheets for students to work their products and record their answers. Circulate while students work to monitor student progress and resolve any disagreements that arise.



6.12.3 Activity: Integer War – Multiplication Edition

Set Up:

- Remove all jokers and face cards (Kings, Queens, and Jacks) from the deck.
- With your partner, determine who is the dealer. The dealer deals the entire deck of cards to each player, face down.

Rules:

- The black cards (Spades ♠ and Clubs ♣) represent positive values.
- The red cards (Diamonds ♦ and Hearts ♥) represent negative values.
- Aces represent either 1 or -1 , while the other cards represent the positive or negative value indicated on the card.

How to Play:

- To begin, each player will flip over one card from their deck and place it in the center.
- The first player who figures out the product of the two integers keeps both cards and adds them to the bottom of their stack.
 - NOTE: If a player says an incorrect product, the other player responds for a chance to win the hand before the first player can change their answer.
- Repeat this process until time is called or until one player has all of the cards in the deck.



Consider your students when planning partners and groups. Produce partnerships that focus on fluency over speed and comprehension over calculation.



Teachers can also increase difficulty by forming some groups of three to play.

6.12.4 Guided Practice: Finding Products or Quotients

Component Overview: Students find the products and quotients of integers. Consider having students check their work with a partner after completing the expressions. Then debrief with a whole-class discussion to clarify any miscalculations and misconceptions, if needed.



6.12.4 Guided Practice: Finding Products or Quotients

1. Find each product or quotient.

a. $10 \times (-65)$

-650

b. -19×99

-1,881

c. $-812 \div (-7)$

116

d. $\frac{208}{-16}$

-13

e. $-125 \div (-5)$

25

f. $(-21)(-16)$

336



Consider allowing students to choose which questions they answer and in what order they answer the questions.



Prompt students to first make sense of what the question is asking before calculating an answer. Provide an example for question 1.

- “The opposite of 65 groups of 10 is what number?”
- “The sign of the product will be negative.”
- “65 groups of 10 would be 650. So, in other words, 650 divided by 65 is 10.”

Students can also decide on the sign of the answer before performing the calculation to use as a way to check the reasonableness of their answers.

6.12.5 Your Turn!

Component Overview: Students work independently to find the products and quotients of integers.



6.12.5 Your Turn!

1. Find each product or quotient.

a. $-17 \times 29 = -493$

b. $41 \cdot (-19) = -779$

c. $-184 \div 23 = -8$

d. $\frac{-175}{-5} = 35$

e. $96 \div (-6) = -16$

f. $(-51) \times (-31) = 1,581$

g. $-256 \div (-16) = 16$

h. $(-17) \cdot 704 \cdot (-1) = 11,968$



*How did you determine the sign of the product or quotient?
How can you check your answer to ensure it is correct?*

6.12.6 Wrap-Up: Ticket Out the Door

Component Overview: Students demonstrate understanding of the lesson learning target by filling in missing factors, divisors, dividends, and quotients in equations. Use data from this formative assessment to determine which students need remediation before the next lesson. Students should not use a calculator to complete the Wrap-Up.



6.12.6 Wrap-Up: Ticket Out the Door

- Complete the table by writing a value in each blank that completes each related equation.

Multiplication Equation	...related to...	Division Equation
$(-12)(-42) = 504$	$\longleftrightarrow$	$504 \div (-42) = -12$
$-7 \times 18 = -126$	$\longleftrightarrow$	$-126 \div 18 = -7$
$-50 \cdot 13 = -650$	$\longleftrightarrow$	$-650 \div 13 = -50$



Notice the focus of this Wrap-Up is centered on the relationship between multiplication and division rather than the calculation. Consider this when planning your instruction throughout the lesson.